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Q) An environmental research team wants to automatically group different flower species based on their physical characteristics. the dataset iris.csv contains the following attributes: SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm, and species.

tasks:

1. load the iris dataset and display the first few records.
2. drop the species column and consider only numerical features for clustering
3. apply k means clustering with n_clusters = 3 (since the dataset has three species)
4. add the cluster labels to the dataset as a new column named Cluster
5. visualize the clusters using a scatter plot (eg. PetalLengthCm vs PetalWidthCm) with different colours for ech cluster
6. compare the obtained cluster labels with the actual species column to evaluate clustering performance

1. Import libraries and load the dataset

```
In [17]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
import warnings
warnings.filterwarnings("ignore")

df = pd.read_csv(r"C:\Users\adwai\Downloads\Iriss.csv")

print(df.head())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

2. Select numerical features

```
In [18]: X = df[[
    "SepalLengthCm",
    "SepalWidthCm",
    "PetalLengthCm",
    "PetalWidthCm"
]]

print(X.head())
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

3. Apply K-Means clustering

```
In [19]: kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

df["Cluster"] = kmeans.fit_predict(X)

print(df["Cluster"])
```

0	1
1	1
2	1
3	1
4	1
...	
145	2
146	0
147	2
148	2
149	0

Name: Cluster, Length: 150, dtype: int32

4. Add cluster labels

```
In [20]: print(df.head())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
es \						
0	1	5.1	3.5	1.4	0.2	Iris-seto
sa						
1	2	4.9	3.0	1.4	0.2	Iris-seto
sa						
2	3	4.7	3.2	1.3	0.2	Iris-seto
sa						
3	4	4.6	3.1	1.5	0.2	Iris-seto
sa						
4	5	5.0	3.6	1.4	0.2	Iris-seto
sa						

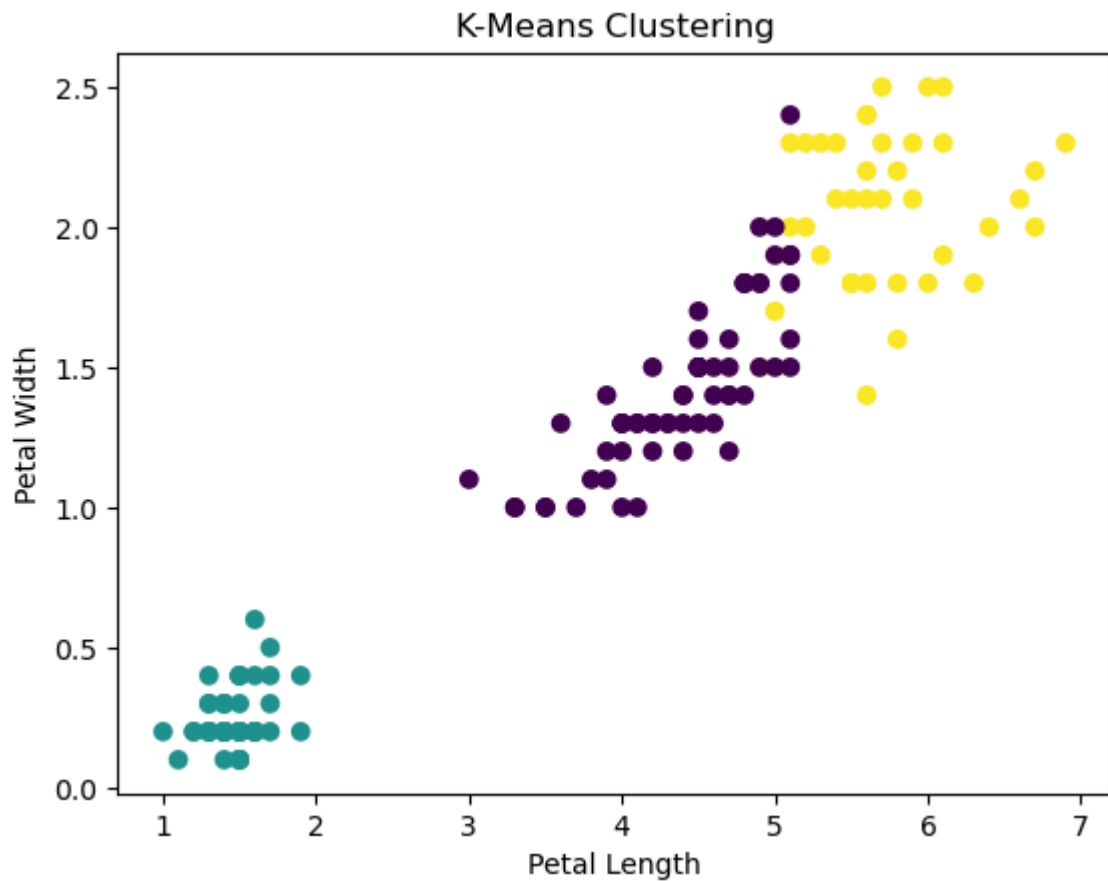
	Cluster
0	1
1	1
2	1
3	1
4	1

5. Visualize clusters

```
In [21]: plt.scatter(
            df["PetalLengthCm"],
            df["PetalWidthCm"],
            c=df["Cluster"]
        )

plt.xlabel("Petal Length")
plt.ylabel("Petal Width")
plt.title("K-Means Clustering")

plt.show()
```



6. Compare clusters with actual species

```
In [22]: print(df[["Species", "Cluster"]])

print(pd.crosstab(df["Species"], df["Cluster"]))
```

	Species	Cluster
0	Iris-setosa	1
1	Iris-setosa	1
2	Iris-setosa	1
3	Iris-setosa	1
4	Iris-setosa	1
..	...	...
145	Iris-virginica	2
146	Iris-virginica	0
147	Iris-virginica	2
148	Iris-virginica	2
149	Iris-virginica	0

[150 rows x 2 columns]

Cluster	0	1	2
Species			
Iris-setosa	0	50	0
Iris-versicolor	48	0	2
Iris-virginica	14	0	36

7. Evaluate clustering performance

```
In [23]: from sklearn.metrics import accuracy_score
```

```
mapping = {}

for cluster in df["Cluster"].unique():
    mapping[cluster] = df[df["Cluster"] == cluster]["Species"].mode()[0]

df["PredictedSpecies"] = df["Cluster"].map(mapping)

accuracy = accuracy_score(
    df["Species"],
    df["PredictedSpecies"]
)

print("Accuracy:", accuracy * 100, "%")
```

Accuracy: 89.33333333333333 %

In []: